



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 101: Calculus I

First Exam, 07 March 2019

Duration: 90 minutes

- *In questions 1-8, show all your work.*
- *Calculators and communication devices are not allowed in the examination room.*

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (25 pts)	
Q2 (15 pts)	
Q3 (5 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (10 pts)	
Q7 (10 pts)	
Q8 (15 pts)	
Total	

1. $[3 + 3 + 3 + 3 + 3 + 5 + 5 = 25 \text{ pts.}]$ Let

$$f(x) = \begin{cases} e^x - \frac{3}{x-7} + 5, & \text{if } x \leq 1, \\ \ln(x-1) + 3, & \text{if } 1 < x \leq 3 \\ \frac{x^2 + 1}{x^3 + 1}, & \text{if } x > 3. \end{cases}$$

(a) Find $\lim_{x \rightarrow 1} f(x)$, if it exists.

(b) Find $\lim_{x \rightarrow 3} f(x)$, if it exists.

(c) Find $\lim_{x \rightarrow -\infty} f(x)$, if it exists.

(d) Find $\lim_{x \rightarrow \infty} f(x)$, if it exists.

(e) Find, if any, all vertical asymptotes of the graph of f .

(f) Find and classify, if any, all the points of discontinuity.

(g) Find, if any, all horizontal asymptotes of the graph of f .

2. $[5 + 5 + 5 = 15 \text{ pts.}]$ Let $f(x) = \frac{x+1}{2x+1}$.

(a) Show that f is one-to-one.

(b) Find the inverse function of f .

(c) Find the domain and range of f .

3. $[5 \text{ pts.}]$ Find the exact value of $\sin \left(\cos^{-1} \left(\frac{4}{5} \right) \right)$.

4. [10 pts.] Solve the equation $\ln(x^2 - 2x - 2) = 0$.

5. [10 pts.] If $e^x \leq f(x) \leq \frac{4x^2 + 1}{x^2}$, find $\lim_{x \rightarrow 0} x^4 f(x)$, if it exists.

6. [10 pts.] Find the values of a and b , if any, that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & \text{if } x < 2, \\ ax^2 - bx + 4, & \text{if } 2 \leq x < 3 \\ 2x - a + b, & \text{if } x \geq 3. \end{cases}$$

7. [10 pts.] Use the **definition of the derivative** to find $f'(3)$ of $f(x) = \sqrt{x - 2} + 3$.

8. [5 + 10 = 15 pts.] a) State the Intermediate Value Theorem.

b) Show that there is at least one real root of the equation $\cos(\sqrt{x}) = e^x - 2$.

1. [3 + 3 + 3 + 3 + 3 + 5 + 5 = 25 pts.] Let

$$f(x) = \begin{cases} e^x - \frac{3}{x-7} + 5, & \text{if } x \leq 1, \\ \ln(x-1) + 3, & \text{if } 1 < x \leq 3 \\ \frac{x^2+1}{x^3+1}, & \text{if } x > 3. \end{cases}$$

(a) Find $\lim_{x \rightarrow 1} f(x)$, if it exists.

We have $\lim_{x \rightarrow 1^+} f(x) = -\infty$, so $\lim_{x \rightarrow 1} f(x)$ does not exist.

(b) Find $\lim_{x \rightarrow 3} f(x)$, if it exists.

We have $\lim_{x \rightarrow 3^+} f(x) = \frac{10}{28}$ and $\lim_{x \rightarrow 3^-} f(x) = \ln 2 + 3$, so $\lim_{x \rightarrow 3} f(x)$ does not exist.

(c) Find $\lim_{x \rightarrow -\infty} f(x)$, if it exists.

We have $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} e^x - \frac{3}{x-7} + 5 = 5$.

(d) Find $\lim_{x \rightarrow \infty} f(x)$, if it exists.

We have $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{x^2+1}{x^3+1} = 0$.

(e) Find, if any, all vertical asymptotes of the graph of f .

The only vertical asymptote of the graph of f is $x = 1$ because $\lim_{x \rightarrow 1^+} f(x) = -\infty$.

(f) Find and classify, if any, all the points of discontinuity.

The function has infinite discontinuity at $x = 1$ and jump discontinuity at $x = 3$.

(g) Find, if any, all horizontal asymptotes of the graph of f .

The graph of f has two horizontal asymptotes $y = 0, 5$.

2. [5 + 5 + 5 = 15 pts.] Let $f(x) = \frac{x+1}{2x+1}$.

(a) Show that f is one-to-one.

(b) Find the inverse function of f .

(c) Find the domain and range of f .

If $f(x_1) = f(x_2)$, then this implies that $\frac{x_1+1}{2x_1+1} = \frac{x_2+1}{2x_2+1}$ and this is equivalent to say that $2x_1x_2 + 2x_2 + x_1 + 1 = 2x_1x_2 + x_2 + 2x_1 + 1$. It is clear that the last statement implies that $x_1 = x_2$. Therefore, f is one-to-one. To find the inverse we let $y = \frac{x+1}{2x+1}$ and this is equivalent to say that $2xy + y = x + 1$ or $x = \frac{1-y}{2y-1}$. Therefore, the inverse function is given by $f^{-1}(x) = \frac{1-x}{2x-1}$. The domain of f is $\mathbb{R} \setminus \{-\frac{1}{2}\}$ and the image is $\mathbb{R} \setminus \{\frac{1}{2}\}$.

3. [5 pts.] Find the exact value of $\sin \left(\cos^{-1} \left(\frac{4}{5} \right) \right)$.

We let $x = \cos^{-1} \left(\frac{4}{5} \right)$, then $\cos x = \frac{4}{5}$ and therefore we obtain $\sin \left(\cos^{-1} \left(\frac{4}{5} \right) \right) = \sin x = \frac{3}{5}$.

4. [10 pts.] Solve the equation $\ln(x^2 - 2x - 2) = 0$.

We have $x^2 - 2x - 2 = 1$ after we take both sides to base e . This last equation is equivalent to $x^2 - 2x - 3 = 0$ or $(x - 3)(x + 1) = 0$. Therefore, the roots for this equation are $x = -1$ and $x = 3$.

5. [10 pts.] If $e^x \leq f(x) \leq \frac{4x^2 + 1}{x^2}$, find $\lim_{x \rightarrow 0} x^4 f(x)$, if it exists.

We multiply the given inequality by x^4 to obtain $x^4 e^x \leq x^4 f(x) \leq \frac{4x^6 + x^4}{x^2}$. Now we have $\lim_{x \rightarrow 0} x^4 e^x = 0$, $\lim_{x \rightarrow 0} \frac{4x^6 + x^4}{x^2} = \lim_{x \rightarrow 0} 4x^4 + x^2 = 0$. As a result of Squeeze Theorem, we obtain $\lim_{x \rightarrow 0} x^4 f(x) = 0$.

6. [10 pts.] Find the values of a and b , if any, that make f continuous everywhere.

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & \text{if } x < 2, \\ ax^2 - bx + 4, & \text{if } 2 \leq x < 3 \\ 2x - a + b, & \text{if } x \geq 3. \end{cases}$$

The function has to be continuous at $x = 2$ and at $x = 3$. Therefore, we obtain $f(2) = \lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$ and $f(3) = \lim_{x \rightarrow 3^+} f(x) = \lim_{x \rightarrow 3^-} f(x)$. This implies $4a - 2b + 4 = 4$ and $6 - a + b = 9a - 3b + 4$. Hence, we get $a = 1$ and $b = 2$.

7. [10 pts.] Use the **definition of the derivative** to find $f'(3)$ of $f(x) = \sqrt{x - 2} + 3$.

We have

$$\begin{aligned} f'(3) &= \lim_{h \rightarrow 0} \frac{f(3+h) - f(3)}{h} = \lim_{h \rightarrow 0} \frac{\sqrt{3+h-2} + 3 - \sqrt{3-2} - 3}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sqrt{3+h-2} - 1}{h} = \lim_{h \rightarrow 0} \frac{(3+h-2) - 1}{h(\sqrt{3+h-2} + 1)} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{3+h-2} + 1} = \frac{1}{2} \end{aligned}$$

8. [5 + 10 pts.] a) State the Intermediate Value Theorem.

b) Show that there is at least one real root of the equation $\cos(\sqrt{x}) = e^x - 2$.

a) Suppose f is a continuous function on $[a, b]$ with N between $f(a)$ and $f(b)$. Then there exists $c \in (a, b)$ such that $f(c) = N$.

b) The function $f(x) = \cos(\sqrt{x}) - e^x + 2$ has domain $[0, \infty]$. Now we have $f(0) = 1 - 1 + 2 > 0$ and $f(2) = \cos(\sqrt{2}) - e^2 + 2 < 0$. The function f is continuous as it is a linear combination of continuous functions. Hence by the IVT, there exists a number c in $(0, 2)$ such that $f(c) = 0$. This is equivalent to say that $\cos(\sqrt{c}) - e^c + 2 = 0$ or $\cos(\sqrt{c}) = e^c - 2$. Therefore, c is a real root of the given equation.